

Perpetual Futures and Cryptocurrency Market Quality

Evidence from Exchange Closures and Staggered Introductions

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Motivation

- ▶ Perpetual futures dominate crypto derivatives (93% of volume)
- ▶ Daily BTC perp volume: \$180B (6x NVIDIA)
- ▶ **Key question:** How do perpetual futures affect spot market quality?
- ▶ Two natural experiments:
 1. Huobi exchange closure (Oct 2021, China ban)
 2. Staggered introduction across 38 coin-exchange pairs

Data & Sample

- ▶ **Source:** Kaiko consolidated order book (Level 10)
- ▶ **Exchanges:** Binance, Binance Futures, Huobi, Huobi DM, OKEx
- ▶ **Coins:** 13 major USDT pairs (BTC, ETH, SOL, ADA, etc.)
- ▶ **Period:** 2017–2023 (67 months max)
- ▶ **Microstructure metrics:** Quoted spread, effective spread, depth, volatility, OFI, Kyle's λ

Key Finding 1: Spot vs. Perpetual Spread

	Spot	Perpetual
Spread (%)	3.20	1.50
Depth (\$M)	2.09	4.96

[Figure: spread_ratio_heatmap.png]

- ▶ Perpetual spread 53% tighter
- ▶ Perpetual depth 2.4x larger
- ▶ Universal across all 13 coins

Key Finding 2: Huobi DiD

Setup: Huobi terminated perpetual futures Oct 28, 2021 (China ban)

Result: DiD = +0.0026 ($p = 0.017$)

- ▶ Spot spread *widened* after perpetual termination
- ▶ Robust across 9 specifications
- ▶ Effect concentrated in December ($p = 0.001$) and low-volatility days ($p = 0.005$)
- ▶ Inventory channel: $|OBI| \times \text{DiD coefficient} = -0.086$ ($t = -10.16$)

[Figure: event_study_parallel_trends.png]

Key Finding 3: Staggered Introduction

Setup: 38 coin-exchange pairs introduced perpetual futures at different dates on Binance

	Pre	Post	Change
Spread	0.2303	0.1279	-44.5%***
Depth (\$)	584,788	1,852,745	+216.8%***

- ▶ Spread narrows 44.5% after perpetual introduction
- ▶ Depth increases 217%
- ▶ Consistent with information-based market microstructure theory

Microstructure Deep Dive

5,213 daily observations

13 pairs $\times$ 3 exchanges

- ▶ Quoted spread: 0.03%
- ▶ Effective spread: $\sim 1\%$
- ▶ Gap $\Rightarrow$ most trades execute outside best quotes
- ▶ Huobi depth: \$1.6M
- ▶ Binance depth: \$14.8M

[Figure: microstructure_metrics.png]

Robustness

1. **Parallel trends:** No significant pre-treatment coefficients
2. **Placebo tests:** Negative anticipation Aug–Sep, flips at Oct 28
3. **TWFE:** DiD = +0.0013 ($p = 0.028$)
4. **Wild cluster bootstrap:** SE = 0.0029, $p = 0.38$ (13 clusters)
5. **Winsorization 1%:** DiD = +0.0035 ($p = 0.0004$)
6. **Window sensitivity:** All 7 lengths significant ($p < 0.005$)
7. **Heterogeneity:** Late (Dec) and calm (low-vol) effect

Conclusion

- ▶ Perpetual futures **improve** spot market quality
- ▶ Natural experiment: termination $\Rightarrow$ wider spreads, lower depth
- ▶ Mechanism: Information channel (adverse selection) + Inventory rebalancing
- ▶ Policy implication: Banning derivatives may *harm* underlying markets

Thank you

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